

Image Manipulation Techniques and Visual Effects

Presentation & Term Project

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The schedule in the following weeks

- ▶ **11th week: 5/7 (Thurs.), video**
 - ▶ **Lecture**
- ▶ **5/7(Thurs.) ~ 5/14 (Thurs.)**
 - ▶ **Finding** your **team members** and **registering** at the **online sheet** listed on E3New (before 4pm, 5/14).
 - ▶ **1~4** members per group.
 - ▶ We encourage you to team up; **2~3** members per team are appropriate.
 - ▶ If you need help to find teammates, please follow the instruction posted by TA (or send him an email).
 - ▶ If you want to form a **4-member** team, there should be **more** components in your project or the project scope should be broader.
- ▶ **12th week: 5/14 (Thurs.), video**
 - ▶ **Lecture** (dependent on the lecture progress)

The schedule (cont.)

- ▶ **13th week: 5/21 (Thurs.)**
 - ▶ Preparing presentation video.
 - ▶ (no in-person class)
- ▶ **Before 23:59, 5/25 (Mon.)**
 - ▶ **Submit your team presentation video: project survey and proposal.**
 - ▶ 1-person team: 8~16 minutes.
 - ▶ 2-or 3-person team: 12~20 minutes.
 - ▶ 4-person team: 16~25 minutes.
 - ▶ In the video,
 - ▶ List your *project title, member info.*
 - ▶ Talk about the **goal** of your project, **survey of related work**, and your **plan** (such as expected methods, the difficulties/uniqueness, experiments, etc.)
*(Allocate more time to the part of **survey**, and focus on one or two related methods/techniques)*

The schedule (cont.)

- ▶ **14th ~15th week: 5/26 (Tues.) ~ 6/8 (Mon.)**
 - ▶ Follow TA's instruction, **browse** a number of **videos**, and **provide commends** or **ask questions**.
 - ▶ (no in-person class)
- ▶ **16th week: 6/11 (Thur.)**
 - ▶ Final presentation and demo I (physical/in-person, with a tiny bonus)
 - ▶ The presentation time depends on the team amount.
 - ▶ About the **goal** of your project, **related work/techniques**, **your system and methods**, **the difficulties/uniqueness**, **results and demo**.
 - (Allocate more time to the last 3 sections, especially **results and demo**)*
- ▶ **17th week: 6/18 (Thur.)**
 - ▶ Final presentation and demo II (online, e.g. Google Meet)
 - ▶ (presentation through video only for special circumstances)
 - ▶ The time and the rule are similar to those in demo I.

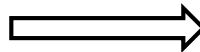
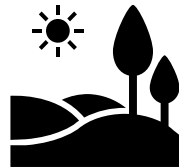
The schedule (cont.)

▶ Before 23:59, 6/22 (Mon.)

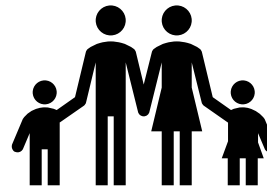
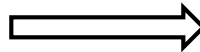
- ▶ Following the TAs' announcement and uploading your **term project report** (no more than 6 pages) and **codes** (with some comments)
 - ▶ Describe the goal/problem, method, difficulty and/or uniqueness, results and/or comparison.
 - ▶ (Must) List libraries and open sources used in your project.
 - ▶ (Must) What are the differences between your project and open-source/others' projects you used.
 - ▶ List the contribution (e.g. work items) of each member?
 - ▶ If there is any *confidential* problem, please discuss the issue with Teacher in advance.

Term project: possible directions

- ▶ Developing a tool for specific image-editing tasks.
 - ▶ You may integrate multiple components (some implemented by your team and some open sources) into one framework.
 - ▶ E.g. A building inserting tool.
 - ▶ Image matting + inpainting + color adjustment (+ shadowing)



- ▶ E.g. A multi-source compositing tool with editing functions.



Term project: possible directions

- ▶ Implement one or more interesting image manipulation work from references.
 - ▶ In addition to reproduction, it would be great if you can adjust something, improve it, or find something new.
 - ▶ E.g. Image editing with generators

Note: you may need more than 24G VRAMs for high-resolution StyleGAN training

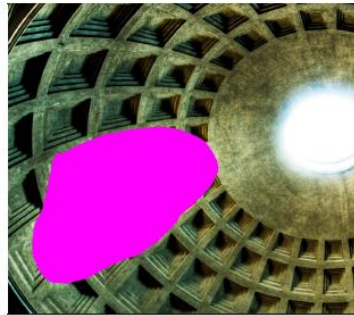


Term project: possible directions

- E.g. image inpainting or reshuffling.



Barnes et al. “PatchMatch: A Randomized Correspondence Algorithm for Structural Image Editing”, ACM Trans. Graphics (also in Prof. SIGGRAPH’09).



Darabi et al., “Image Melding: Combining Inconsistent Images using Patch-based Synthesis”, ACM Trans. Graphics (Prof. SIGGRAPH’12).

Term project: possible directions

- E.g. image inpainting.



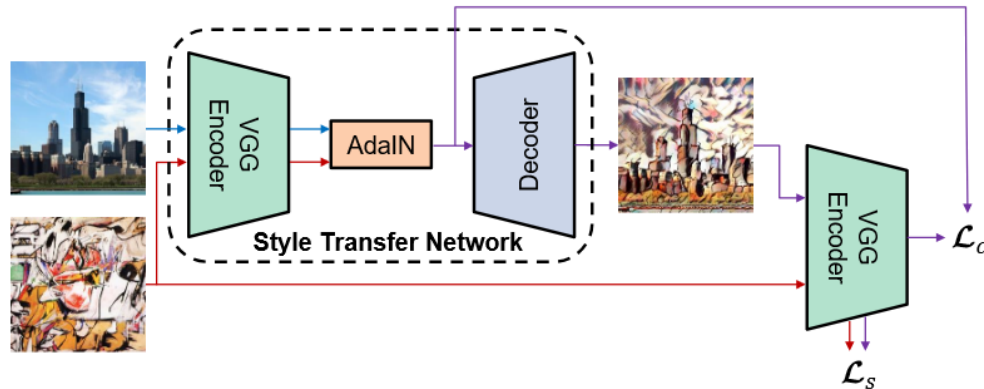
Yu et al., Free-Form Image Inpainting with Gated Convolution, ICCV'19.



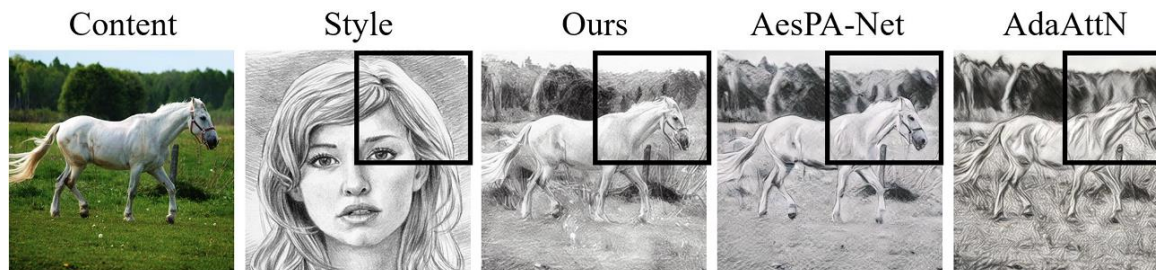
Lugmayr et al., RePaint: Inpainting using Denoising Diffusion Probabilistic Models, CVPR'22.

Term project: possible directions

► E.g. style transfer



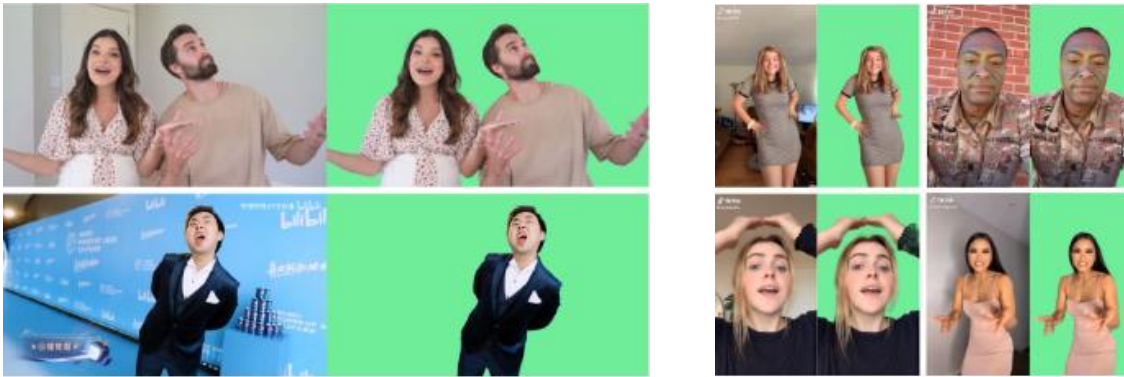
Huang and Belongie, Arbitrary Style Transfer in Real-time with Adaptive Instance Normalization, ICCV'17.



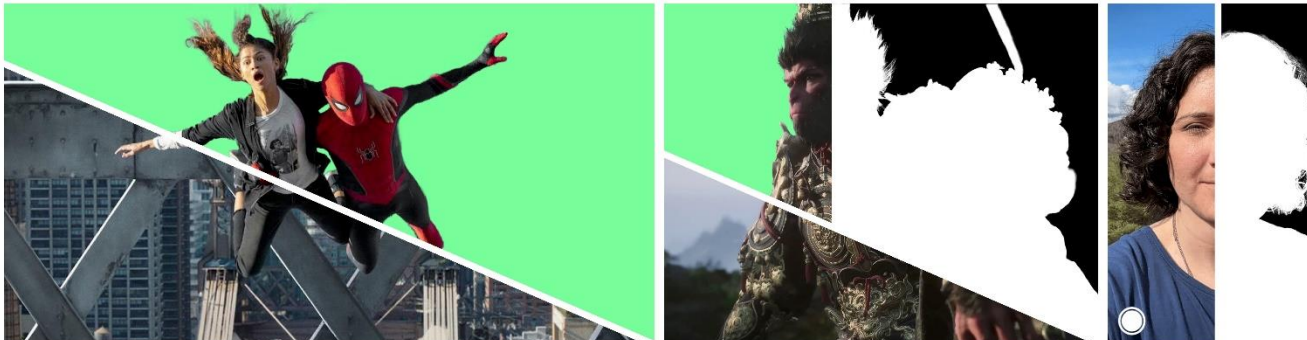
Chung et al., Style Injection in Diffusion, CVPR'24.

Term project: possible directions

- ▶ E.g. image or video matting



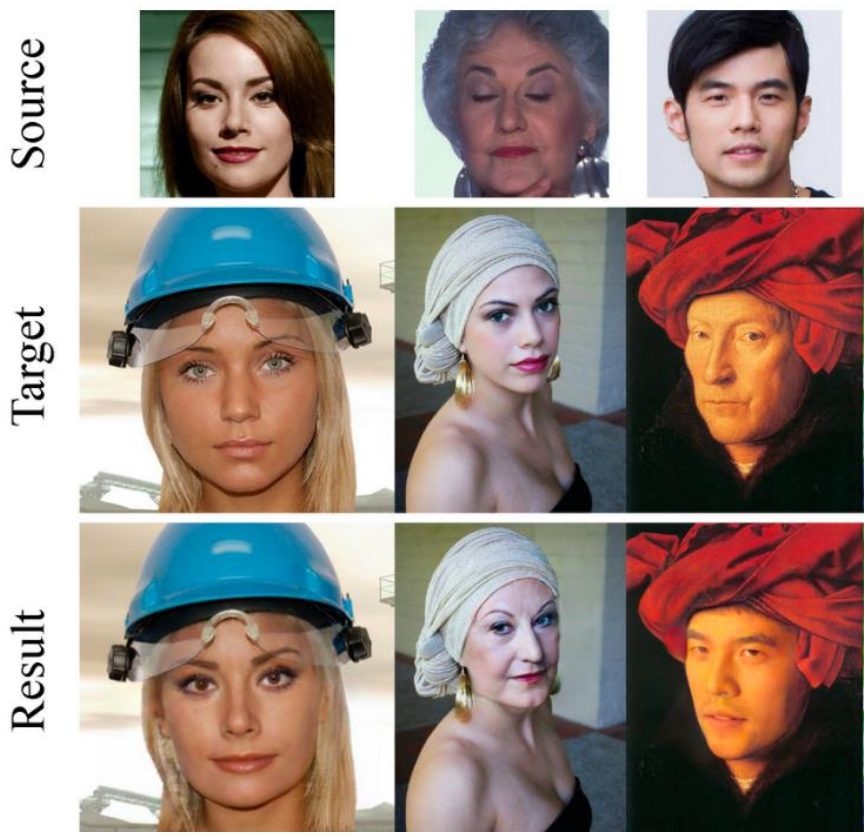
Lin et al., Robust High-Resolution Video Matting with Temporal Guidance, WACV'22.



Yang et al., MatAnyone, CVPR'25.

Term project: possible directions

► E.g. face swapping



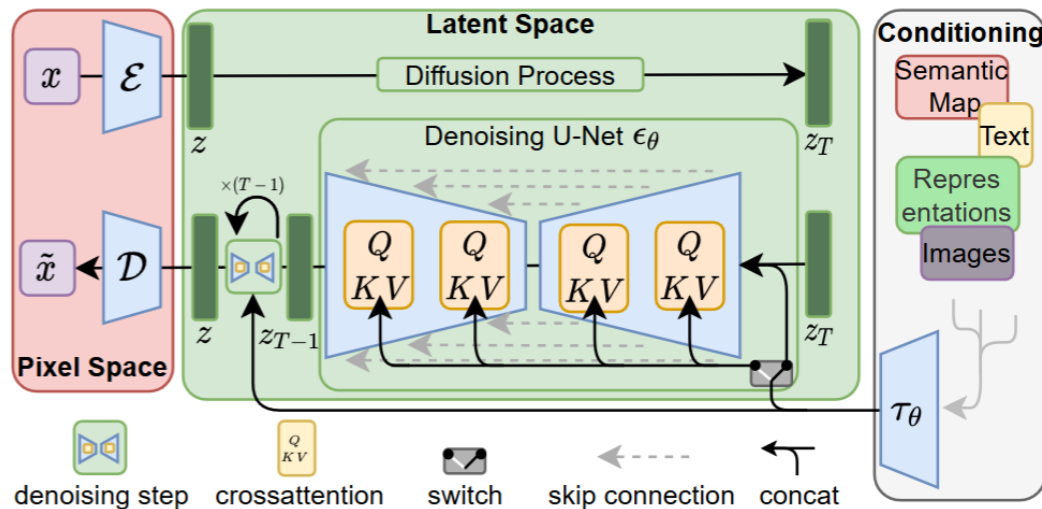
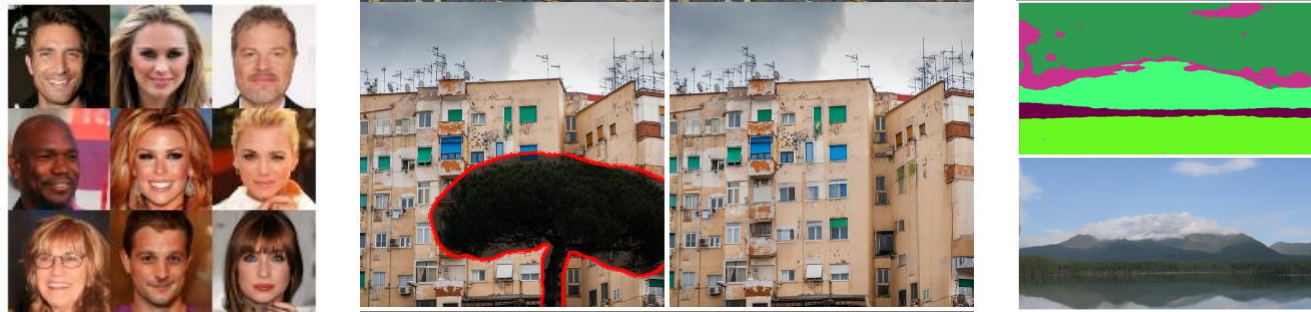
Li et al., (FaceShifter) Advancing High Fidelity Identity Swapping for Forgery Detection, CVPR'20.



Li et al., 3D-Aware Face Swapping, CVPR'23

Term project: possible directions

- E.g. applying modern latent diffusion models.

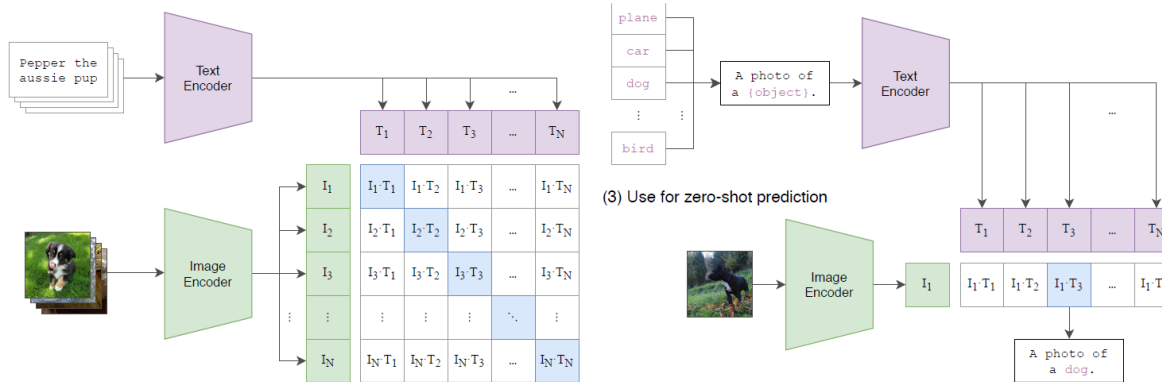


Note: you may need more than 24 or 32 GB VRAMs to train a high-resolution LDM model.

Term project: possible directions

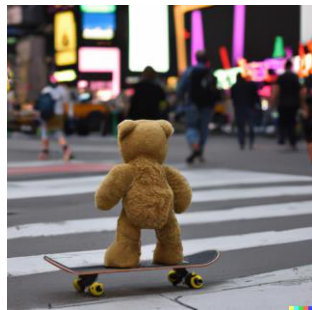
► E.g. work with modern text-vs-image frameworks

CLIP

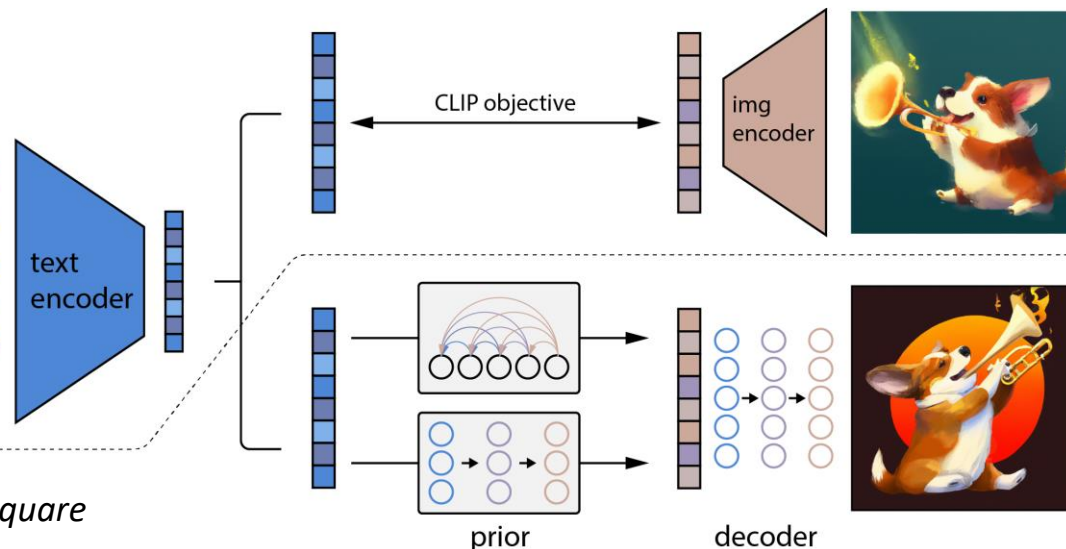


Radford et al., Learning Transferable Visual Models From Natural Language Supervision, ICML'21

DALL-E 2



"a corgi playing a flame throwing trumpet"



a teddy bear on a skateboard in times square

Ramesh et al., Hierarchical Text-Conditional Image Generation with CLIP Latents, arxiv 2022.

References

- ▶ Proc. Intl. Conf. Computer Vision and Pattern Recognition (CVPR)
- ▶ Proc. Intl. Conf. Computer Vision (ICCV)
- ▶ Proc. Euro. Conf. Computer Vision (ECCV)

- ▶ Proc. ACM SIGGRAPH (or ACM Trans. Graphics special issue on July/Aug.)
- ▶ Proc. ACM SIGGRAPH Asia (or ACM Trans. Graphics special issue on Nov./Dec.)

- ▶ Proc. Neural Information Processing Systems (NeurIPS)

- ▶ Proc. ACM Multimedia

- ▶

Choosing full papers (usually >6 pages)

References

- ▶ IEEE Trans. on Pattern Analysis and Machine Intelligence (TPAMI)
- ▶ International Journal of Computer Vision (IJCV)
- ▶ ACM Trans. on Graphics (TOG)
- ▶ IEEE Trans. on Visualization and Computer Graphics (TVCG)
- ▶ IEEE Trans. on Image Processing (TIP)
- ▶ IEEE Trans. on Multimedia (TMM)
- ▶